

STAT 6910 Advanced R

Reproducible Research (RR)

by

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1 Basics about L^AT_EX via RStudio

First, notice that you can open a L^AT_EX .tex file from within RStudio. This file is recognized as a .tex file — and you can translate it via the Compile PDF option from within RStudio!

When working with .tex in RStudio, you have options to introduce some L^AT_EX formatting via highlighting and mouse clicks. Let's try this for the words **bold**, *italic*, and `typewriter`.

Now, we want a new subsection

Notice what is being produced here!

1.1 Now, we want a new subsection

Do this again — but remove the * after \subsection.

And what about a bullet list hereafter?

- AAA
 - BBBB
1. xya
 2. bbbb

AAA This is it

XY Whatever...

1.2 Math Symbols

Let's try a few math symbols: σ , λ , $s_1 = \sum_{i=1}^n i^2$, ∞ , $\sqrt{x}$, $\bar{x}$ and a $\odot$.

Now, do the sum in a math environment:

$$s_1 = \sum_{i=1}^n i^2$$

A few errors: $s_{11} = \sum_{i=1}^n i^2$
\$

100%

Compare ... with ...

2 L^AT_EX Documentation

Numerous web sites provide information about L^AT_EX, including tutorials and help pages. Here are some:

- <http://www.latex-project.org/>
- <http://en.wikibooks.org/wiki/LaTeX>
- <http://miktex.org/>
- <http://www.artofproblemsolving.com/Wiki/index.php/LaTeX:Symbols>

If you need help on a specific latex topic, google for `latex` and the keyword or command you are interested in.

3 Using Sweave and knitr in RStudio

Detailed information can be found at <http://www.rstudio.com/ide/docs/authoring/overview>.

Basically, you need to start at **File** – **>** **New** – **>** **R Sweave**. Let's do so.

If you want to use an existing `.tex` file, you must change the file extension to `.Rnw` and add the line

```
\SweaveOpts{concordance=TRUE}
```

after the

```
\begin{document}
```

line.